

TOMATOAI

GREENHOUSE INTELLIGENCE PLATFORM

Motivation

- Need for technology in agriculture
- Manual greenhouse monitoring





Goal

- Build an AI-powered smart platform
- Support plant monitoring & greenhouse management

Problems

01 Delayed and Inaccurate Disease Diagnosis

02 Lack of Supply Chain Transparency

03 Slow Response to Incidents





Solutions

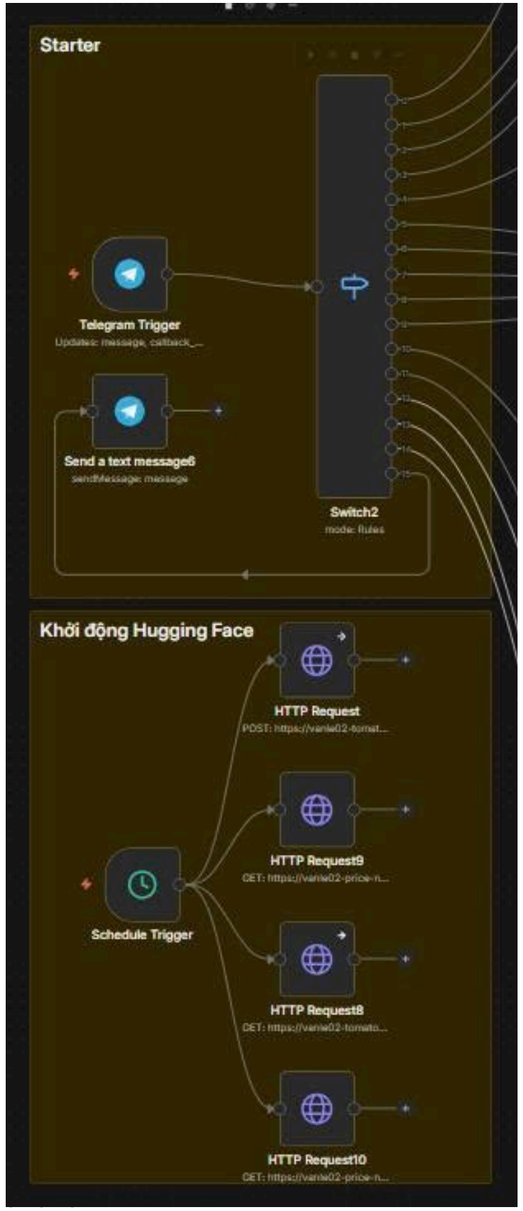
01 AI Disease Diagnosis

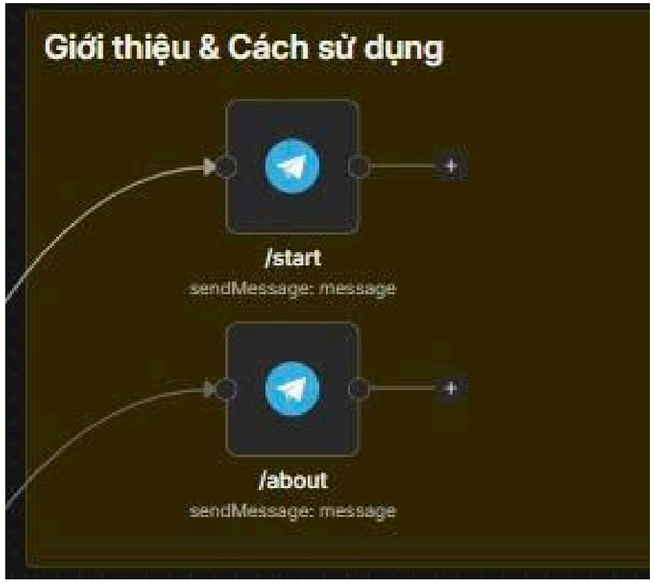
02 Smart Harvest Gradin

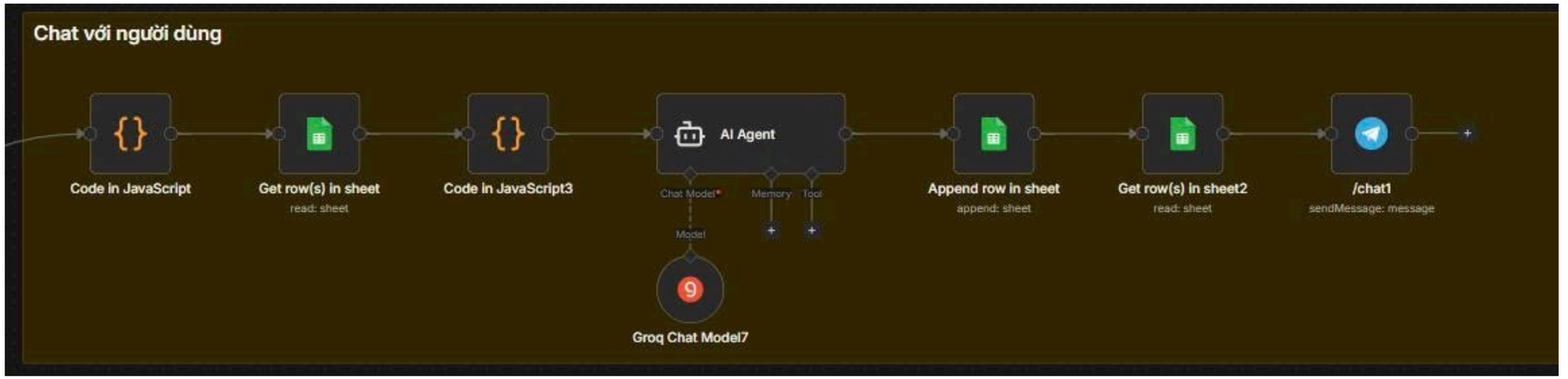
03 Traceability

04 Instant Alerts

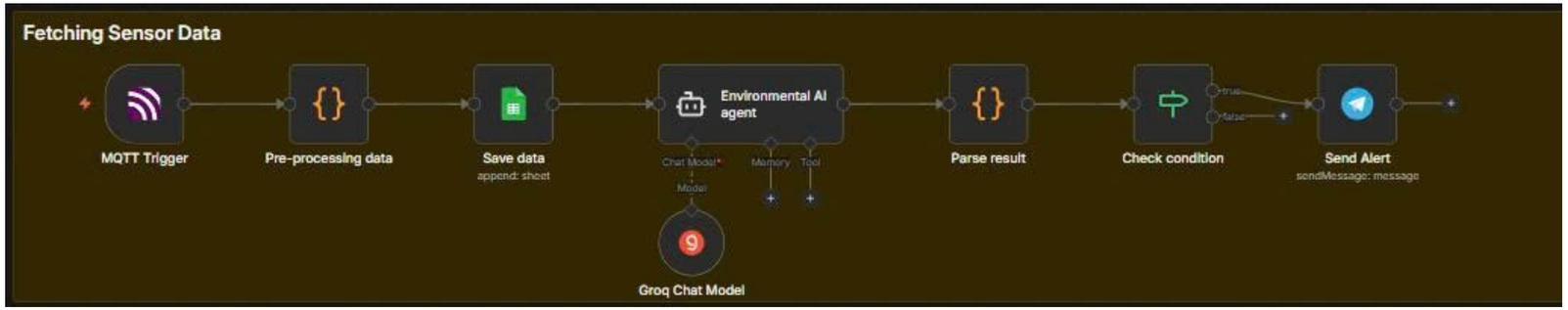
Platform

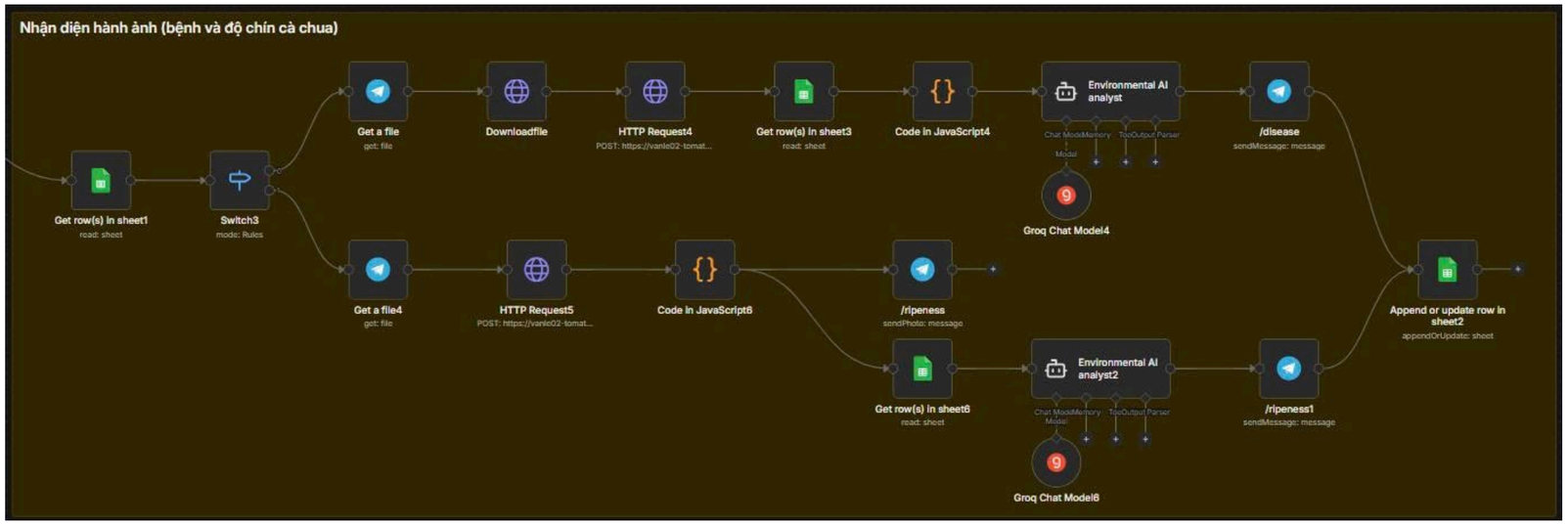

Starter
Telegram Trigger
Send a text message6
sendMessage: message
Switch2
mode: Rules
Khởi động Hugging Face
HTTP Request
HTTP Request9
Schedule Trigger
HTTP Request8
HTTP Request10

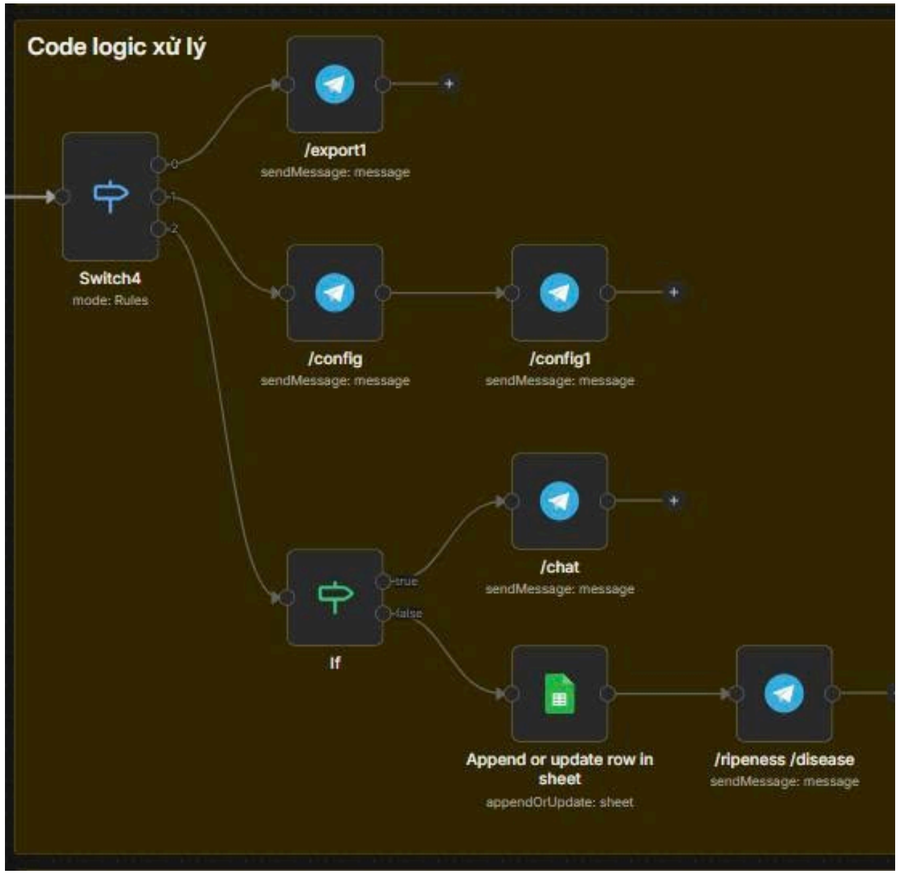

Giới thiệu & Cách sử dụng
/start
sendMessage: message
/about
sendMessage: message


Chat với người dùng
Code in JavaScript
Get row(s) in sheet
read: sheet
Code in JavaScript3
AI Agent
Chat Model*
Memory
Tool
Model
Groq Chat Model7
Append row in sheet
append: sheet
Get row(s) in sheet2
read: sheet
/chat1
sendMessage: message

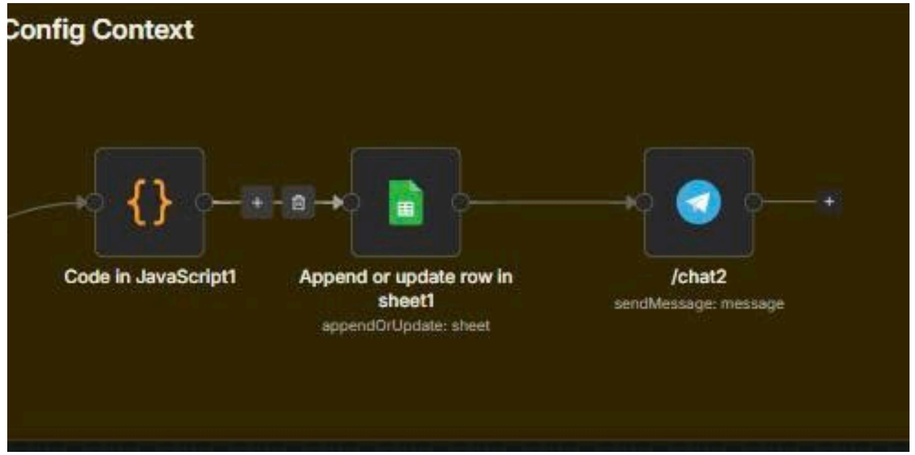
Technical:

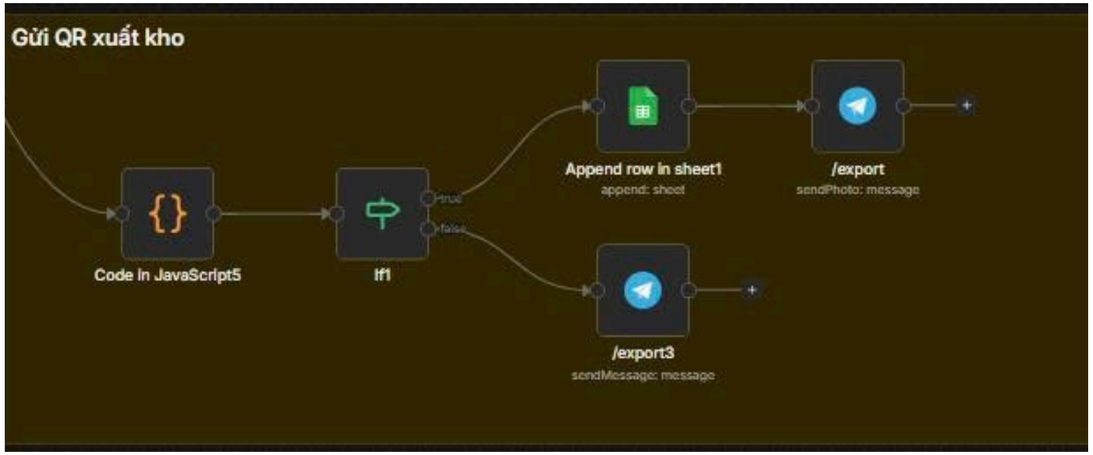

Fetching Sensor Data
MQTT Trigger
Pre-processing data
Save data
append: sheet
Environmental AI agent
Chat Model*
Memory
Tool
Model
Groq Chat Model
Parse result
Check condition
true
false
Send Alert
sendMessage: message

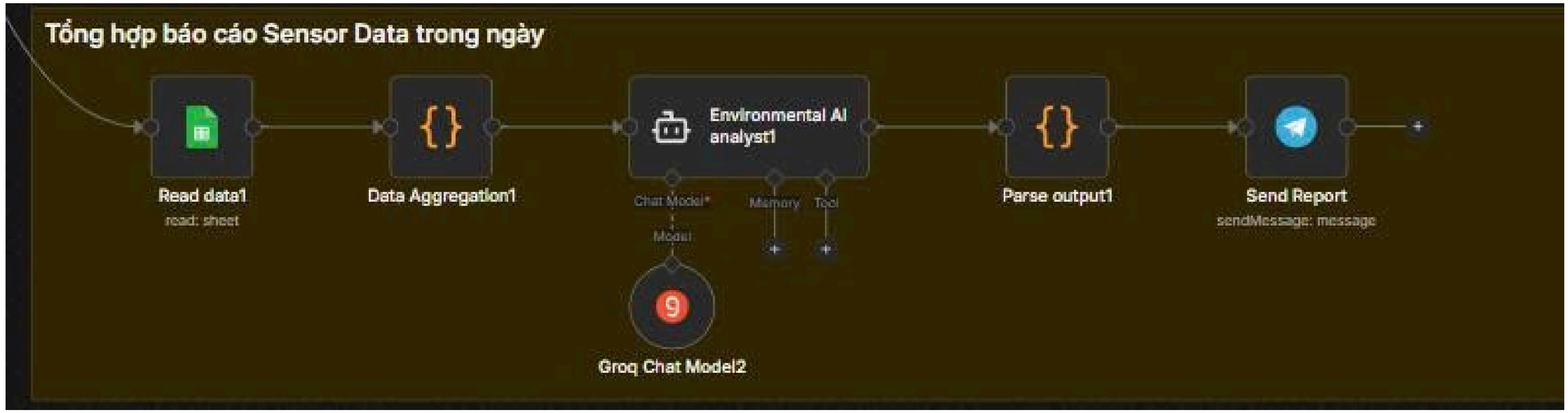

Nhận diện hành ảnh (bệnh và độ chín cà chua)
Get row(s) in sheet1
read: sheet
Switch3
mode: Rules
Get a file
get: file
Downloadfile
HTTP Request4
POST: https://vanle02-tomat...
Get row(s) in sheet3
read: sheet
Code in JavaScript4
Environmental AI analyst
Chat Model
Memory
Tool
Output Parser
Model
Groq Chat Model4
/disease
sendMessage: message
Get a file4
get: file
HTTP Request5
POST: https://vanle02-tomat...
Code in JavaScript6
/ripeness
sendPhoto: message
Get row(s) in sheet6
read: sheet
Environmental AI analyst2
Chat Model
Memory
Tool
Output Parser
Model
Groq Chat Model6
/ripeness1
sendMessage: message
Append or update row in sheet2
appendOrUpdate: sheet


Code logic xử lý
Switch4
mode: Rules
/export1
sendMessage: message
/config
sendMessage: message
/config1
sendMessage: message
If
true
false
/chat
sendMessage: message
Append or update row in sheet
appendOrUpdate: sheet
/ripeness /disease
sendMessage: message

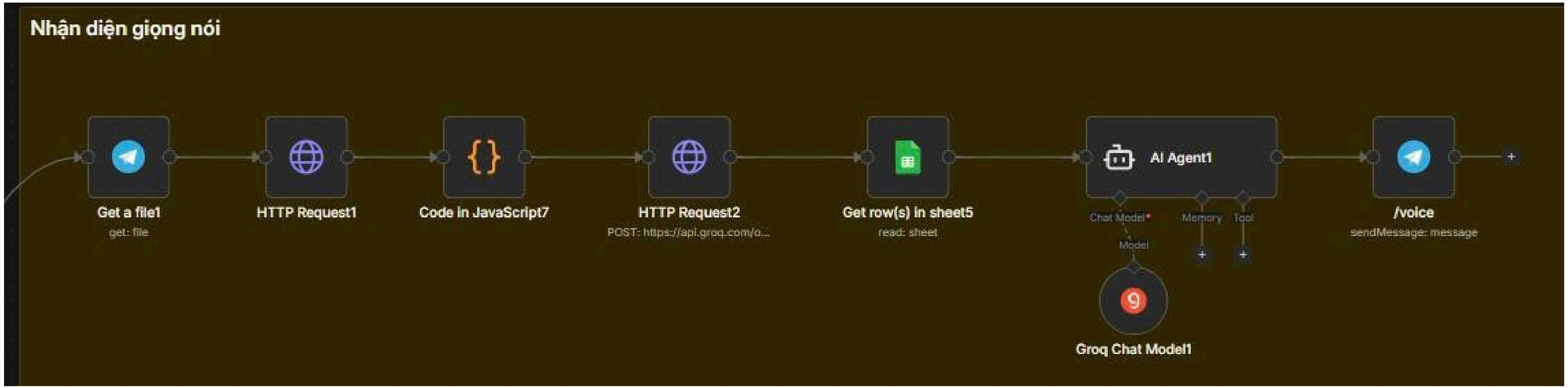
Technical:

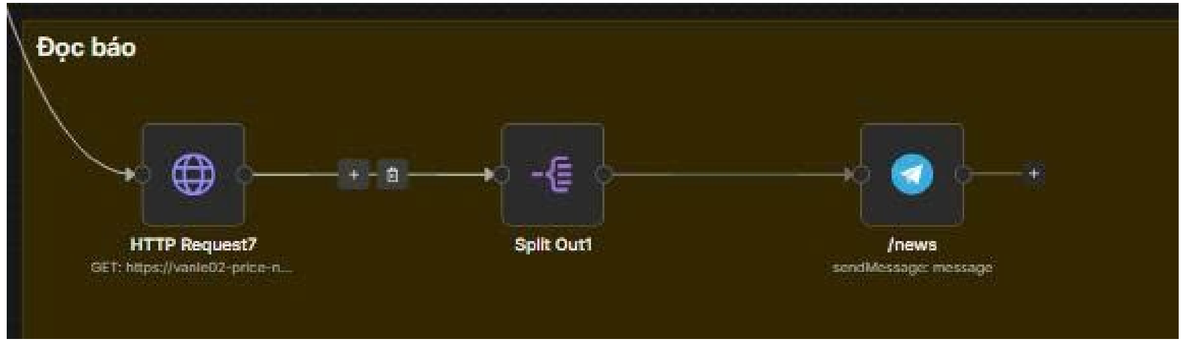

Config Context
Code in JavaScript1
Append or update row in sheet1
appendOrUpdate: sheet
/chat2
sendMessage: message

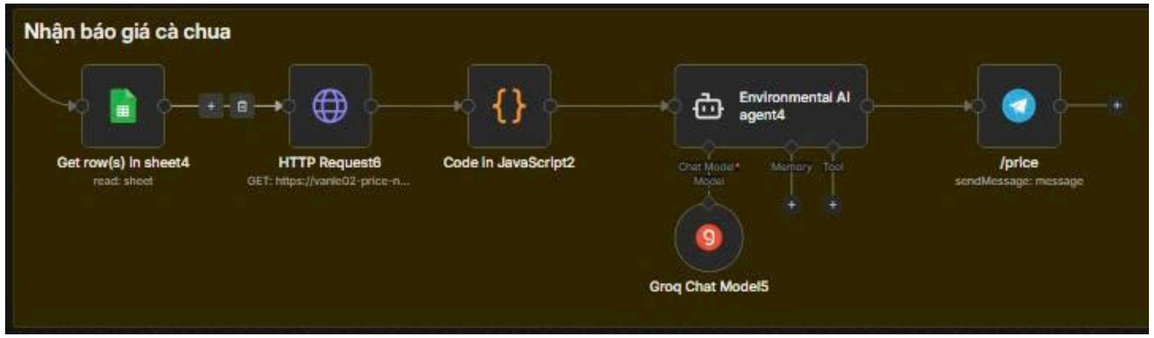

Gửi QR xuất kho
Code in JavaScript5
If1
true
false
Append row in sheet1
append: sheet
/export
sendPhoto: message
/export3
sendMessage: message


Tổng hợp báo cáo Sensor Data trong ngày
Read data1
read: sheet
Data Aggregation1
Environmental AI analyst1
Chat Model*
Memory
Tool
Model
Groq Chat Model2
Parse output1
Send Report
sendMessage: message

System Optimization

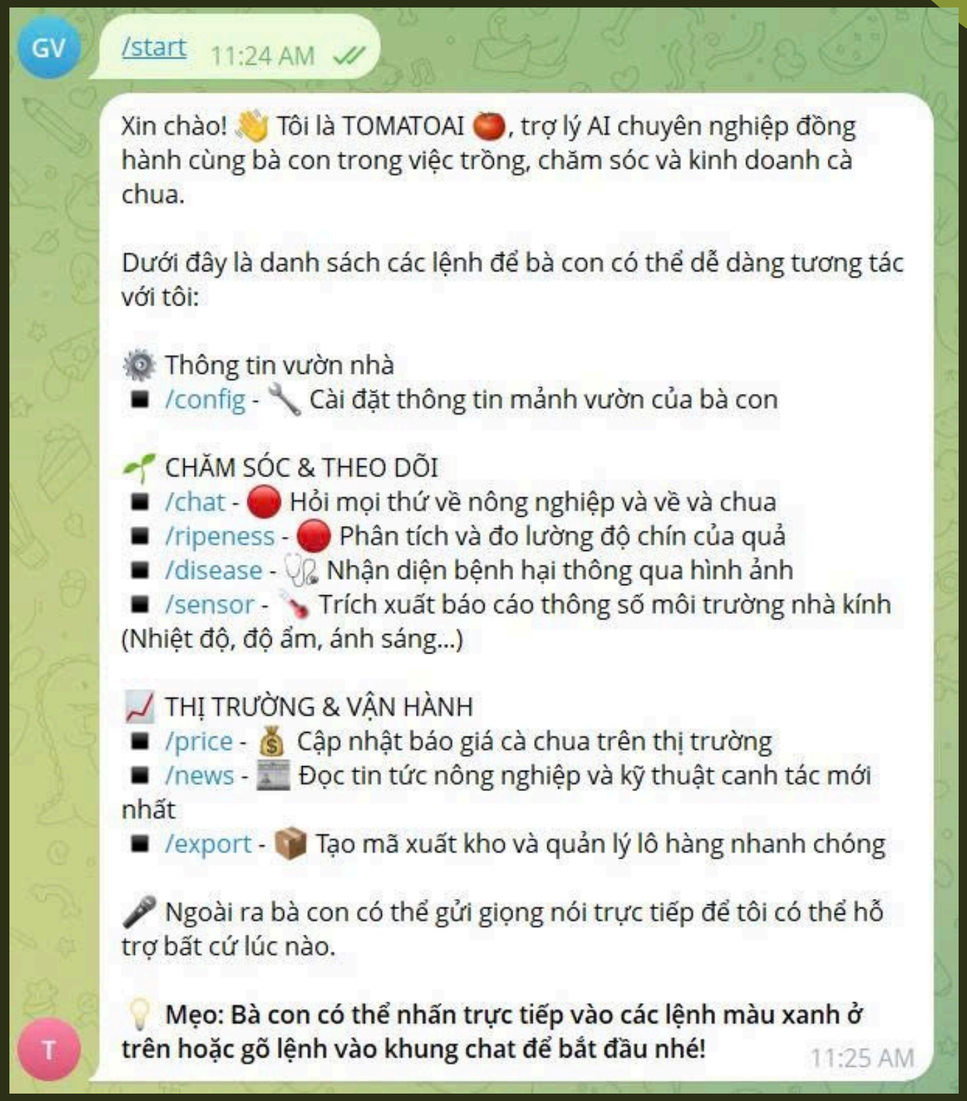






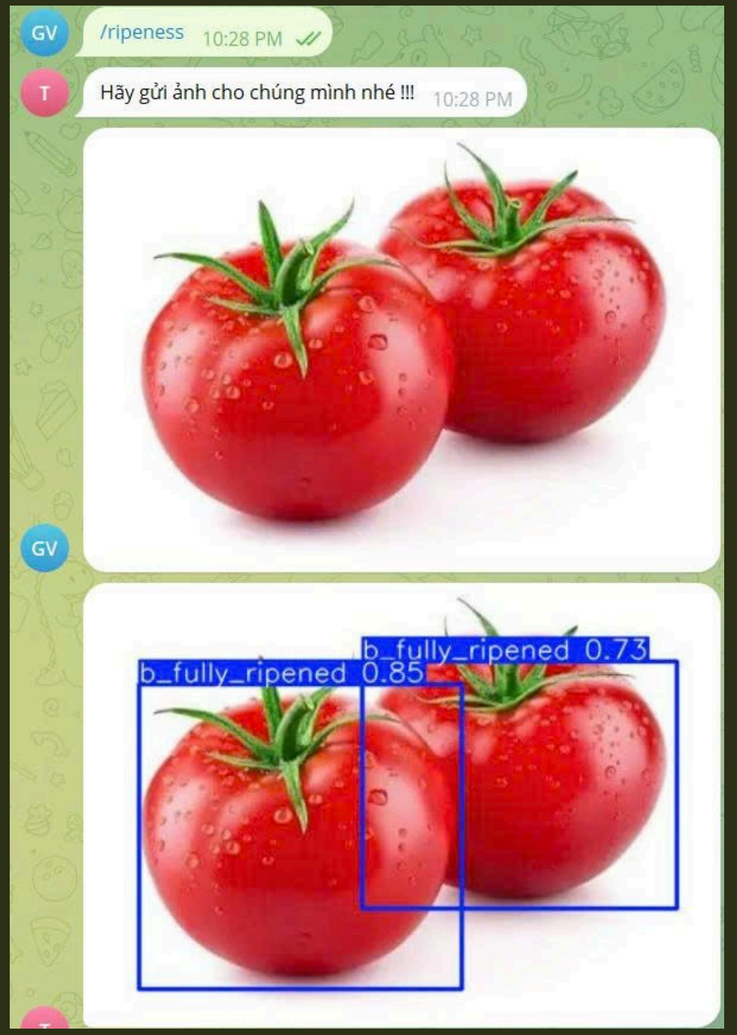
Demo Results





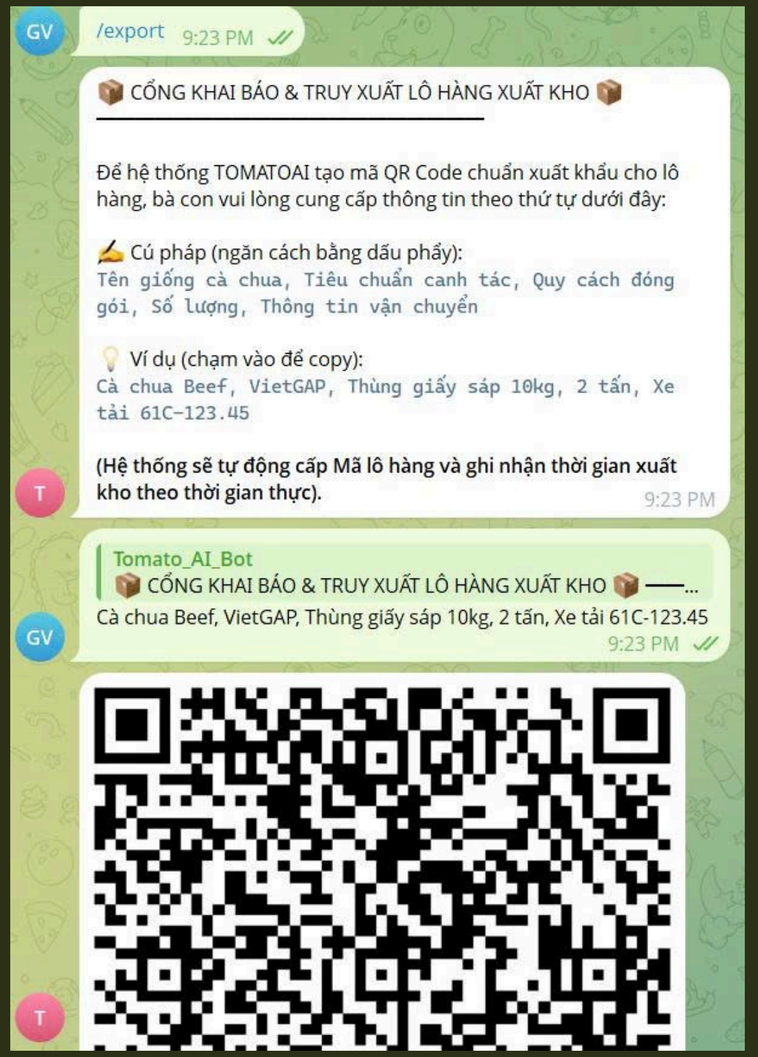
Demo Results





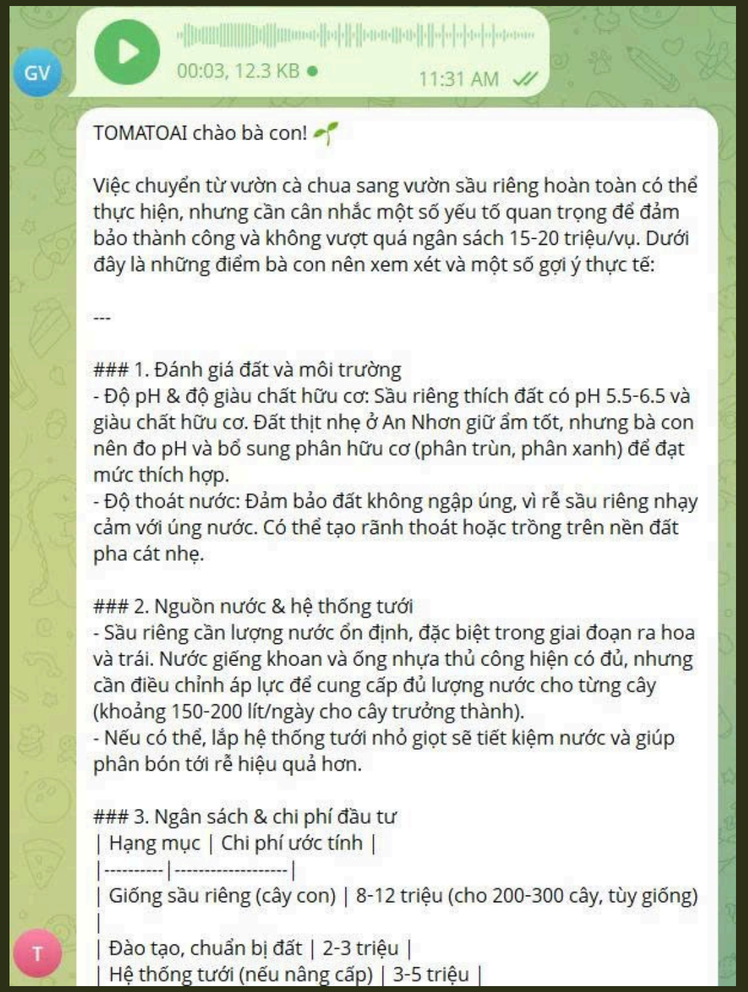
Demo Results





Demo Results





Demo Results

